



P24

**Locating the RSI Spectrum in
Practice: L1–L5 Laddering of
RSIAgent, Ouroboros,
Dream-RSI, Gödel Agent,
HyperAgents, and RSIAI0 against
the Theseus Criteria**

RSI 谱系实测定位：L1–L5 判别标准



Locating the RSI Spectrum in Practice: L1–L5 Laddering of RSIAgent, Ouroboros, Dream-RSI, Gödel Agent, HyperAgents, and RSIAI0 against the Theseus Criteria

Working Title (EN): Locating the RSI Spectrum in Practice: L1–L5 Laddering of RSIAgent, Ouroboros, Dream-RSI, Gödel Agent, HyperAgents, and RSIAI0 against the Theseus Criteria

Mid-volume theory paper P24 • connecting v7.5.0 P14's RSI five-level ladder with Theseus Labs' seven criteria • placing six public 2026 RSI systems one by one • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: "recursive self-improvement (RSI)" is now used too loosely—as long as a system "gets better with use" someone calls it RSI. Theseus Labs (arXiv:2609.11873) gives a strict ruler: genuine RSI must meet seven criteria (cross-round learning, persistent retention, system self-modification, candidate

proposal, update verification, successor re-entry, mechanism revision). This paper uses this ruler to place RSIAgent, Ouroboros, Dream-RSI, Gödel Agent, HyperAgents, RSIAI0 six systems one by one on the L1–L5 ladder. **Conclusion first: none reaches L5; RSIAgent and Ouroboros are L2, HyperAgents closest to L5 but not there.**

Evidence caliber (stated once for the whole paper): this is a concept-location paper, **containing no new UDOS measured numbers.** Six external systems' existence and core mechanisms verified online 2026-09-21 (arXiv:2609.11873 / 2609.14858 / ACL2025 long.1354 / 2603.19461 / Zenodo 15644670 / 2609.15364); performance numbers (e.g. "317 calls $\approx$ 51200 generations") all paper/official reports, not independently rechecked, this paper not using them to support level judgments, level judgments only based on the structural fact "which layer the system modified." UDOS ladder follows v7.5.0 P14/P15.

Abstract

"RSI" has become a marketing buzzword. To judge how far a system actually went, one needs a ruler not carried away by performance numbers. Theseus Labs' "Toward Genuine Recursive Self-Improvement" (arXiv:2609.11873) proposes the Headroom-Closed Index (HCI) quantifying existing LLMs' RSI gap, and lists seven criteria genuine RSI must meet: cross-round learning, persistent retention, system self-modification, candidate proposal, update verification, successor re-entry, mechanism revision. This paper connects this ruler with UDOS v7.5.0 P14/P15's five-level ladder (L1 modify weights / L2 modify Harness / L3 modify architecture / L4 search priors / L5 structural self-modification), placing six public 2026 systems one by one. Key judgments: RSIAgent (arXiv:2609.15364) only modifies memory, lands **L2**; Ouroboros (arXiv:2608.08311) modifies reviewed Harness, lands **L2**; Dream-RSI (arXiv:2609.14858) does not modify weights, only optimizes exploration strategy, lands **between L3/L4 toward L4** (improving "exploration strategy structure," not network connections); Gödel Agent (ACL2025 long. 1354) self-modifies source and logic, lands **L4**; HyperAgents (arXiv:2603.19461) makes the meta-improvement process itself modifiable, the **public system closest to L5**, but this paper judges it **not yet L5**—because "meta-layer editable" does not equal "the system has stably and recursively modified the rules determining subsequent modifications," lacking successor re-entry closed-loop evidence. RSIAI0 (Zenodo 15644670) is an L2–L4 open-source experimental bench. This paper's core claim: **all 2026 public evidence still concentrated in L2–L4, L5 (structural self-modification) has no controlled public achievement**; any claim directly asserting "gets better with use" as L5 RSI should be checked item by item against the seven criteria.

Keywords: recursive self-improvement; RSI criteria; HCI; L1–L5 ladder; structural self-modification; HyperAgents; Dream-RSI; Gödel Agent

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** RSI used too broadly, "gets better with use" and "genuine recursive self-improvement" conflated, causing systematically optimistic self-improvement progress judgments.
- **Core argument:** dual-layer location using Theseus seven criteria + UDOS L1–L5 ladder; the judgment standard is "which structural layer the system modified," not how much performance improved.

- **Evidence lines:** (1) Theseus seven criteria and HCI (arXiv:2609.11873, existence verified); (2) six systems' mechanism descriptions (all existence verified); (3) P14/P15's five-level ladder and "recent RSI = optimizing how answers are obtained" conclusion.
 - **Contribution:** six-system one-by-one grading table; explicit "HyperAgents closest to L5 but not there" judgment; a discipline "before claiming L5 must pass seven checks."
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1 Problem: RSI Used Too Loosely

1.1 Three "Self-Improvements" Conflated

In the market "RSI" mixes at least three things:

1. **Regular training/fine-tuning:** modify weights, structure unchanged (L1). This is not RSI, it is normal training.
2. **Modify Harness/memory:** modify tools, prompts, context management, memory stores, not model weights (L2). This is already "the system modifying its own running scaffolding."
3. **Modify structure itself:** modify network connections, optimizers, "rules determining subsequent modifications" (L3–L5). This is the RSI truly needing vigilance and audit.

Calling 1 RSI is concept-swapping; directly inflating 2 into 3 is exaggeration. This paper uses a hard ruler to separate them.

1.2 Theseus' Seven Criteria

Theseus Labs (arXiv:2609.11873) gives seven items genuine RSI should simultaneously meet (this paper restates concepts per the ledger, exact wording per the original): cross-round learning, persistent retention, system self-modification, candidate proposal, update verification, successor re-entry, mechanism revision. Plain-language version:

- **Cross-round learning:** this round's improvement truly affects the next round, not isolated temporary tuning;
- **Persistent retention:** improvement not erased by next restart;
- **System self-modification:** modifying the system itself, not the external environment;
- **Candidate proposal:** improvements appearing as comparable "candidate schemes";
- **Update verification:** verification after changes, not trusting immediately;
- **Successor re-entry:** the changed system can continue as the "object being changed," forming a recursive loop;
- **Mechanism revision:** highest order—the system can modify "the modification mechanism itself."

HCI (Headroom-Closed Index) is its metric quantifying RSI gaps, this paper cites no specific values, only borrowing the framework "RSI is a graded, classifiable, unclosed process."

RSI L1-L5 阶梯与六系统落点



Figure 1 L1 modify weights → L5 structural self-modification five-level ladder, overlaid with six-system placements (concept location, not performance ranking). Conceptual schematic, not measured data.

2 Placing Six Systems One by One

Judgment only looks at "which layer modified," not performance gains.

2.1 RSI Agent → L2

RSI Agent (arXiv:2609.15364) coordinates curriculum/actor/verifier, using BRS+DRS to build the **memory store**. It does not modify model weights, it modifies "the Harness layer of memory." The paper reports open-source models surpassing closed-source on specified benchmarks (paper/media report, not independently rechecked)—but that is Harness-RSI's result, landing **L2**. Key reminder: its memory is **environment-specific** (see P18), not meeting the implicit high-RSI requirement "cross-environment persistent transfer."

2.2 Ouroboros → L2 (review-driven)

Ouroboros (arXiv:2608.08311) improves tools, prompts, context assembly, core implementation through **reviewed commits**. It modifies Harness, each step reviewed. Lands **L2**. Its dross expanded in P19: review bandwidth caps recursion depth, not autonomous L5.

2.3 Dream-RSI → L3/L4 toward L4

Dream-RSI (arXiv:2609.14858, exact title Recursive Self-Improvement through Evolving Worlds) organizes completed discovery history into a replayable simulator, "dreaming in the discovery tree" to improve **exploration strategy**, not underlying weights. It modifies "exploration strategy structure"—one layer above memory (L2), near L4 (search/modify strategy structure), but not network connections themselves. The paper reports "about 317 calls $\approx$ evolutionary search 51200 generations" (official

caliber, not independently rechecked), this paper not upgrading its level by performance. Lands **L4 edge**.

2.4 Gödel Agent → L4

Gödel Agent (ACL2025 long.1354, arXiv:2410.04444), inspired by Gödel machines, can **dynamically modify its own logic and behavior code** from high-level goal prompts alone. It self-modifies source and behavior logic, a clear L4 (modify executable structure) instance. Multiple secondary sources mark it L4, this paper follows, **not calling L5**—because self-modifying code does not equal "modifying the rules themselves determining subsequent modifications."

2.5 HyperAgents → Closest to L5, but Not There

HyperAgents (arXiv:2603.19461, Meta FAIR 2026) merges task Agents and meta Agents into a single editable program, key innovation **the meta-layer modification process itself editable** (metacognitive self-modification). This is the only one of six systems directly touching the L5 criterion "mechanism revision." **But this paper explicitly judges it not yet L5**: HyperAgents demonstrated the **capability** of meta-layer editability, still lacking public controlled evidence of "the system stably, recursively, self-verifyingly modifying the rules determining subsequent modifications" successor re-entry closed loop. It stands at L5's threshold; crossing needs what P14 calls the fifth stage (autonomously modifying the optimizer and stably benefiting).

2.6 RSIAIO → L2–L4 Experimental Bench

RSIAIO (Zenodo records/**15644670**, note not 15646184) is an open-source experimental framework, containing VM controlled execution, memory systems and Darwinian Mode (managing populations of self-modifiable AI variants). It is not a finished product at some level, but an **experimental bench allowing you to walk from L2 to L4**. Positioned as "tool, not landing point."

Table 1 Six-system RSI level location (structural judgment)

System	arXiv / source	Which layer modified	This paper's grade	Meets L5?
RSIAgent	2609.15364	memory (Harness)	L2	no
Ouroboros	2608.08311	reviewed Harness	L2	no
Dream-RSI	2609.14858	exploration strategy structure	L4 edge	no
Gödel Agent	ACL2025 long. 1354	self-modified source/logic	L4	no
HyperAgents	2603.19461	meta-layer modification process editable	threshold • not there	not achieved
RSIAIO	Zenodo 15644670	self-modifiable experimental bench	L2–L4 tool	—

3 Dross / Failure Boundaries / Falsifiable Conditions

3.1 Dross (three traps misreading "gets better with use" as L5)

1. **Performance gain ≠ level gain.** "317 calls $\approx$ 51200 generations," "open-source surpasses closed-source" are efficiency numbers answering "how much faster," not "which layer modified." This paper deliberately does not upgrade levels by performance.

2. **Meta-layer editable $\neq$ already L5.** HyperAgents provides meta-layer editing capability, but L5 needs the successor re-entry **closed loop**—rules changed, new rules recursively changed, and stably benefiting. Lacking this, meta-layer editable is only "can do," not "already doing."
3. **Review-driven $\neq$ autonomous RSI.** Ouroboros-style reviewed commit depth capped by review bandwidth (P19), should not count as L5 autonomous evolution.

3.2 Failure boundaries

- If a system claims L5, must pass Theseus seven criteria item by item; any item (especially "successor re-entry" and "mechanism revision") lacking controlled public evidence, this paper insists on downgrading to L4.
- Level judgments rely on systems' **self-reported mechanism descriptions**; if internal actual modification depth not public, this paper handles as "insufficient evidence, no upward adjustment."

3.3 Falsifiable conditions

- **F24-a:** if within the next 12 months a public, reproducible L5 system appears passing all seven criteria, stably benefiting in independent evaluation, this paper's "2026 no public L5 achievement" judgment must immediately be revised upward—precisely that judgment's value (overturnable by new evidence).
- **F24-b:** if HyperAgents' later versions publish successor re-entry closed-loop evidence (meta-rule self-modification + verification + re-entry + stable benefit), its grade should rise from "threshold" to "L5 candidate achieved."

4 Conclusion

RSI's real progress is far more conservative than marketing narratives. Measured by the dual ruler of Theseus seven criteria + UDOS L1–L5 ladder, 2026's six representative systems all land L2–L4: RSIAgent, Ouroboros L2; Dream-RSI, Gödel Agent L4; HyperAgents stands at L5's threshold but not crossed. This is not disparaging these works—they prove self-improvement at the "modify Harness/strategy/code" level is genuinely feasible and effective—but refusing to directly translate "gets better with use" into "structural self-modification has arrived."

Level looks at which layer modified, not how many times faster. Before public evidence stably passes the seven criteria, L5 remains a declaration at the threshold, not a crossed fact.

References

RSI theory

1. Schmidhuber, J. (2007/2009). Gödel Machines: Fully Self-Referential Optimal Self-Improvers. (Gödel machine theory source)
2. Theseus Labs (2026). The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement. arXiv:2609.11873. (HCI + seven criteria)

2026 frontier materials (existence verified online 2026-09-21; performance numbers paper/official reports, not independently rechecked)

1. Zhu, S., et al. (2026). RSIAgent: Autonomous Exploration for Recursive Self-improvement in New Environments. arXiv:2609.15364.

2. Ouroboros (2026). A Self-Developing Frontier Coding Agent with Reviewed Core Evolution. arXiv: 2608.08311.
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5. Meta FAIR (2026). HyperAgents / DGM-Hyperagents. arXiv:2603.19461.
6. RSIAI0 (2026). Bootstrapping AGI via LLM-Guided Recursive Self-Modification. Zenodo records/ 15644670.

UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P14 "The Convergence of RSI" (seven-route verification + five-stage optimization chain), P15 "Intelligence as a Cross-Substrate Structural Organization Phenomenon" (RSI five-level ladder: L1 weights/L2 Harness/L3 architecture/L4 search priors/L5 structural self-modification).
2. UDOS v7.6.0 paper volume: P18 "BRS+DRS Two-Stage Memory Flywheel," P19 "Structural Change Review Gate," P33 "v7.6.0 Architecture Master Outline."

Evidence Discipline and Reproduction Notes

- This paper is a **concept-location paper**, no new experiments; level judgments based on "which structural layer the system modified," not performance numbers.
- Six systems' existence and mechanism descriptions verified 2026-09-21; performance numbers (317 calls, open-source surpassing closed-source, etc.) all paper/official reports, not independently rechecked, not serving as grading basis.
- RSIAI0's correct record number Zenodo 15644670 (not 15646184).
- "HyperAgents not L5" is a conservative judgment based on "successor re-entry closed-loop evidence missing," if new evidence appears should revise per F24-b.
- This paper claims no existing system has consciousness; RSI levels are functional structural levels, not consciousness levels.